

Dan Evans

Phone: 07803416393

Email: devans8819@gmail.com

Linkedin [Dan Evans](#)

Summary

Versatile and forward-thinking data leader with extensive experience driving innovation through rapid prototyping and dynamic problem-solving. Specializing in AI-driven solutions and intelligent performance metrics, I excel in leading cross-functional teams to conceive and deliver quick-turnaround innovations. My expertise and passion spans machine vision, deep learning, and advanced data analytics, focusing on early-stage development and optimization of consumer technologies.

I bridge technical and non-technical disciplines to achieve shared goals, combining academic research rigor (peer review, T-process collaboration, experimental validation) with structured agile software release methods. This multifaceted approach ensures innovative yet practically deployable solutions that align with strategic objectives and maintain technical excellence.

Areas of Expertise

Data Mining

- Developed automated data cleansing workflows (Python) for ML pipelines.
- Created custom API connectors for seamless cloud data integration.
- Advocate for DVC and data versioning best practices.
- Deployed event-driven pipelines (Azure Functions) for dynamic data updates.
- Conducted EDA to extract insights from user behavior datasets.

Machine Learning [Computer vision]

- Proficiency in most common Neural Network Architectures and Image Frameworks (e.g., TensorFlow, PyTorch, OpenCV, PIL, scikit-image)
- Utilized GANs to generate synthetic datasets for under-represented cases.
- Implemented LSTMs for real-time sensor data analysis.
- Deployed AR applications via custom hand, face, and gesture recognition models.
- Developed lightweight deep learning models for edge-device deployment.
- Integrated foundation models within Dyson RDD workflow with focus on safety and reliability.

Data Visualization

- Built interactive dashboards (Dash) to visualize real-time sensor data.
- Created web-based Plotly visualizations for cross-team insight sharing.
- Designed quick PyQt5 GUIs for prototyping app experiences.

- Prototyped Jupyter Notebook dashboards with inline interactive widgets.

Leadership

- Guided teams of junior data scientists, overseeing successful data-driven projects.
- Mentored non-team members in statistical and data analysis techniques.
- Aligned data-focused initiatives with broader organizational objectives.
- Fostered innovation through hackathons, community engagement, and collaborative challenges.

Other

- Built and ran a motion-capture setup for AR and sensor validation.
- Setup and ran SEM hair analysis in a Korean beauty pop-up setting.
- Expert in food texture analysis (Stable Micro Systems).
- Qualified UR5 robot arm programmer; ask me about the time I built a robotic chess partner.
- Comfortable building custom rigs and prototypes in research settings, where the goal is often less "clean automation" and more "make this actually work and keep learning."

Computer Vision Contributions

- **User-Centric Metrics:** Developed unique, user-specific performance metrics combining logarithmic regression and precise custom analysis to evaluate abstract concepts such as "good style."
- **Innovative Prototyping:** Comfortable leading rapid, self-directed innovation sprints that produce tangible outcomes. A standout example was designing and implementing a **smart mirror prototype** in a two-week hackathon, integrating a full-pipeline LLM agent with gesture control, real-time image augmentation, and facial analysis capabilities.
- **Skin Analysis Techniques:** Applied advanced deep learning techniques to analyze facial characteristics, including detailed evaluation of skin features such as pores and blemishes, to support product development insights.
- **Style Recommendation Systems:** Conducted exploratory research on image style transfer using latent space representations in diffusion models, leveraging multi-dimensional clustering to enhance style-based recommendations.

Achievements

- **Dyson Corrale Performance Metrics:** Lead engineer and registered inventor of novel lab-based performance metrics for the award-winning Dyson Corrale.
- **Chartered Engineer (IMechE):** Fully chartered with the Institution of Mechanical Engineers (Member ID: 80415022).
- **Patented innovations:** Inventor of multiple (4+) patented (and pending) technologies advancing consumer product innovation, and AR technologies. [GB2608157, GB2608157B]
- **Global Experience:** Demonstrated ability to work internationally, represent the business in media, and travel independently.

Technical Skills / Qualifications

Data & Leadership (7+ years)

- Python, MATLAB, R
- Cloud Platforms: GCP, AWS, Azure
- Agile Framework Leadership
- Technical mentorship

Design & Innovation (6+ years)

- CAD (various platforms)
 - ERP Tools
-

Professional Experience

Lead Machine Learning Engineer – Dyson (2022 – Present)

- Oversight and innovation of data-driven R&D projects, ensuring innovative, robust data and ML solutions.
- Spearhead the development of robust data-driven consumer performance metrics.
- Collaborate with multi-disciplinary teams (embedded, app and design) to rapidly prototype and deploy innovative ideas.
- Driven real ML solutions into real innovative, and patented user facing products (AR and mediapipe).

Senior Research Scientist Manager, Image Analysis – Dyson (June 2021 – 2022)

- Led a team specializing in machine vision, ML, and custom LLM agentic solutions.
- Created novel user-focused performance metrics to accelerate design decisions.
- Balanced project timelines with research exploration to maintain cutting-edge results.

Senior Design Engineer, Product Performance – Dyson (April 2020 – June 2021)

- Designed and managed robust verification test systems for haircare product R&D.
- Delivered comprehensive performance analyses aligned with user feedback and field data.
- Coordinated cross-functional efforts to align engineering objectives with user-centric KPIs.

Performance Verification Engineer – Dyson (May 2017 – April 2020)

- Invented and refined bespoke testing methods to gauge prototype performance.
- Benchmarked and validated methods against real-world user trial data.
- Focused on translating user requirements into quantifiable design metrics.

Extended Break – Volunteer English Teacher in Laos (2016)

- Gained cross-cultural communication skills and adaptability.
- Led English lessons for local students, developing leadership and mentoring skills.

Projects Lead, Syteline Implementation – Oxford Instruments GBS (2014 – 2015)

- Coordinated custom software development (Infor) across multiple global sites.
- Gathered requirements from engineering divisions to tailor ERP solutions.
- Organized remote demonstrations and integrated feedback from global stakeholders.

Mechanical Design Engineer – Oxford Instruments Plasma Technology (June 2011 – 2014)

- Owned full lifecycle of product improvements for semiconductor equipment.
 - Collaborated closely with process, operations, and test teams on leading-edge designs.
 - Managed OEM supplier relationships and oversaw rapid design iterations.
-

Education & Training

- **IMechE Chartered Engineer:** ID 80415022
- **Mechanical Engineering BEng (Hons)** – University of Exeter